

PAULINE TRANGIA DEJOS

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SUMMARY

Computer Science senior with hands-on experience building full-stack web applications and integrating machine learning models into real, deployed systems. Comfortable across the stack — from database design and backend APIs to frontend interfaces — with a growing focus on practical AI integration. Has led and managed teams, shipped production-facing systems, and taken projects from mockup to deployment.

EDUCATION

Visayas State University

Bachelor of Science in Computer Science

Baybay City, Leyte

September 2022 - Present

TECHNICAL SKILLS

- **Backend & Systems:** Supabase, Firebase, MySQL, PostgreSQL
- **AI/ML & Data:** TensorFlow, Keras, Scikit-learn, RAG, pgvector
- **Frontend:** React, Vue, Nuxt.js, Next.js, Flutter, Tailwind CSS
- **Languages:** TypeScript, JavaScript, Python, C#, Dart, HTML/CSS
- **Tools:** Git, Docker, Figma, Jupyter, Streamlit

PROJECTS AND EXPERIENCE

VSULHS Connect System

June 2025 - Present

Full-Stack Developer & AI Engineer

- Developed a full-stack guidance information system using Nuxt.js and Supabase with Row-Level Security (RLS) for secure role-based student data access
- Architected an offline-resilient system using Service Workers and IndexedDB to maintain functionality during network interruptions
- Engineered a RAG-based AI assistant using pgvector and hybrid relational/vector storage for semantic search with embeddings

Oviora: PCOS Screening Tool

Nov 2025 – May 2026

- Built a machine learning web app with dual Random Forest models for symptom-based and clinical PCOS screening using Streamlit
- Achieved 0.96 ROC-AUC and 92.7% accuracy through feature engineering and model optimization
- Engineered domain-specific features including BMI, Waist-Hip Ratio, and Symptom Score to improve predictive performance

Fuzzy RUSLE: Fuzzy Logic-Based Erosion Risk Assessment Using RUSLE Factors

Sept 2025 – Oct 2025

Researcher

- Designed a Mamdani fuzzy inference system using scikit-fuzzy to classify erosion risk from RUSLE soil-loss factors (R, K, LS, C, P)
- Built the inference pipeline across contrasting terrain and vegetation scenarios to ensure interpretable risk predictions

CrackScope

Jan 2025 – May 2025

Computer Vision & Machine Learning Engineer

- Developed a computer vision system using Keras and Python for automated structural crack detection and preventative maintenance
- Leveraged MobileNetV2 with custom classification layers and ImageNet pre-trained weights for binary crack classification
- Designed a severity scoring system combining model confidence and crack-area estimation to generate urgency tiers and remediation recommendations

PawsMatch

Jan 2025 – May 2025

Project Manager & Lead Developer

- Led a multidisciplinary team, delegated tasks, resolved Git conflicts, and delivered milestones under tight deadlines.
- Engineered a pet adoption and surrender management system, overseeing design, implementation, and documentation across all phases with Flutter
- Orchestrated data processing, user authentication, and CRUD operations, significantly enhancing platform stability.
- Developed workflows for managing adoption/surrender application requests, including approval/rejection logic and organization-based permissions.
- Practiced adaptive leadership, balancing technical execution with coordination and quality assurance responsibilities.

GigHub

Jan 2024 – May 2024

Software Developer

- Developed a Window Forms desktop application in C# connecting students with freelance gig opportunities
- Utilized Docker for containerized development and optimized MS SQL Server schema design to improve query performance
- Implemented authentication, CRUD operations, and data processing workflows for gig management

LEADERSHIP AND CONTRIBUTIONS

UI/UX & Frontend Lead — CORAL, USSC Connect

Sept 2025 – Present

VSU Baybay - University Supreme Student Council

- Led frontend development across 2 production university systems — a university-wide attendance and clearance management system (10-person team, led 3 frontend devs) for the Supreme Student Council and an event/attendance management system (4-person team) for university organizations
- Established component architecture and enforced design consistency across both codebases through structured refactors, reducing frontend drift across multiple parallel contributors
- Delivered end-to-end features including dashboard and page layouts, maintenance mode UI, receipt flows, attendance management, and bulk import dialogs — all shipped via heavily reviewed PRs
- Drove responsiveness and accessibility improvements across both products iteratively, owning the full cycle from implementation to merge

UI/UX & Frontend Lead — DormPay

Nov 2025 – Present

VSU Baybay - University Inter-Dormitory Student Body

- Delivered mobile-first responsive improvements across admin, super admin, and dormer-facing pages, with consistent loading/empty states and branding updates
- Built core admin and super admin UI, bulk billing and fine import utilities, and automation-facing interfaces for a campus dormitory billing system